

Crypt Arithmetic Problem

Aim

To solve the **Crypt-Arithmetic Problem** using Python by assigning unique digits to letters such that the arithmetic equation is satisfied.

Algorithm

1. Take the words involved in the crypt-arithmetic puzzle.
2. Identify all unique letters.
3. Assign unique digits (0–9) to each letter.
4. Check that leading letters are not assigned zero.
5. Verify whether the arithmetic equation is satisfied.
6. If satisfied, display the solution.

Python Code

```
from itertools import permutations
```

```
letters = ('S','E','N','D','M','O','R','Y')
```

```
for p in permutations(range(10), 8):
```

```
    S,E,N,D,M,O,R,Y = p
```

```
    # Leading letters cannot be zero
```

```
    if S == 0 or M == 0:
```

```
        continue
```

```
    send = 1000*S + 100*E + 10*N + D
```

```
    more = 1000*M + 100*O + 10*R + E
```

```
    money = 10000*M + 1000*O + 100*N + 10*E + Y
```

```
    if send + more == money:
```

```
print("Solution Found:")  
print("SEND =", send)  
print("MORE =", more)  
print("MONEY =", money)  
break
```

Output

```
--  
Solution Found:  
SEND = 9567  
MORE = 1085  
MONEY = 10652  
--
```

Result

The **Crypt-Arithmetic Problem** was successfully solved using Python. The program assigned unique digits to each letter and produced a valid solution satisfying the given arithmetic equation.